

**Target
selection**

Campaign seed

$$\pi_t \propto n_t^{-2}$$

fixed length-based weights

**Search
initialization**

fresh candidate

repair

new initialization on each attempt

**Generator
mutation**

X	I	Z	Y	I	X
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X	I	Z	Y	Z	X
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local Pauli update (example)

Evolved program (Algorithm 1)

$$\pi_t \propto n_t^{-1.5} (1 + 10b_t)$$

stored cost

near-miss bonus b_t

saved $best[t]$

resume repair

update $best[t]$

conditional reuse on a later attempt

X	I	Z	Y	I	X
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I	X	X	I	Z	Y
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rotate one row
by whole qubits

row rotation added to the existing moves